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# MODULE

## \* Introduction to C -

C is general-purpose programming language developed by Dennis Ritchie in 1972. It is used to create system software and application programs. C language is fast, efficient and forms the base of many programming concepts.

## Features of C -

- Simple and easy to learn
- Portable language
- Fast execution
- Supports structured programming.

Syntax - Means the rules used to write a program correctly.

## Basic structure of C program:-

1. Header file used to include libraries. Example -  
`include <stdio.h>`
2. Function program Execution from `main()`.

3. Variable Declaration used to store values.
4. Statement instructions written in program.
5. Return statement ends execution.

Example :-

```
#include <stdio.h>
int main()
{
    int a = 5;
    printf("%d", a);
    return 0;
}
```

### \* CONDITIONS -

Conditions are used to perform decision making or checking the conditions.

There are classifications as follow-

#### 1. If statement :-

Syntax -

```
If (condition) {
    statements;
}
```

2. If - else statement -  
Syntax -

```
If (condition) {  
    statements;  
}  
else  
{  
    statements;  
}
```

3. Switch statement  
Syntax -

```
Switch (variable) {  
    case & value : statements;  
        break;  
    default : statement;  
}
```

4. Nested if  
Used when one if statement is  
inside another.

Syntax -

```
If (condition 1) {  
    If (condition 2) {  
        statements;  
    }  
    else
```

{  
statements;  
}

## \* LOOPS :-

Loops are the used to repeat statements.

Loops are categorised as -

### 1. for loop

Syntax -

for (Initialization; condition;  
increment)

{  
statements;  
}

### 2. while loop

Syntax -

while (condition)

{  
statements;  
}

### 3 do-while loop Syntax-

```
do { statements; }  
while (condition);
```

### \* Programming Problems

#### 1. Square

|   |   |   |   |   |
|---|---|---|---|---|
| * | * | * | * | * |
| * | * | * | * | * |
| * | * | * | * | * |
| * | * | * | * | * |
| * | * | * | * | * |

```
#include <stdio.h>  
int main() {  
    int x, k;  
    for (x = 1; x < 6; x++) {  
        for (k = 1; k < 6; k++) {  
            printf(" * ");  
        }  
        printf("\n");  
    }  
    return 0;  
}
```

EXPLORER

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> HELLO.C

> TIMELINE

← →

hello.c

🔍

Welcome

C: program.c X

C: program.c > main()

1 #include <stdio.h>

2 int main()

3 {

4     int

5         k,

6         x;

7     for (x = 1; x < 6; x++)

8     {

9         for (k = 1; k < 6; k++)

10         {

11             printf("\*");

12         }

13         printf("\n");

14     }

15     return 0;

16 }

17

PROBLEMS

OUTPUT

DEBUG CONSOLE

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PORTS

/Users/macbook/hello.c/program + ▾ 🗑 ⋮ 🔍 ✕

macbook@macbooks-MacBook-Pro hello.c %

macbook@macbooks-MacBook-Pro hello.c % /Users/macbook/hello.c/program

\*\*\*\*\*

\*\*\*\*\*

\*\*\*\*\*

\*\*\*\*\*

macbook@macbooks-MacBook-Pro hello.c %

Ln 5, Col 9

Spaces: 4

UTF-8

LF

{ }

C

Mac

2. Hollow square

```
* * * * *
*
*
*
*
* * * * *
```

```
#include <stdio.h>
```

```
int main () {
```

```
    int i, k, x;
```

```
    for (x=1; x<6; x++)
```

```
    {
```

```
        for (k=1; k<6; k++)
```

```
        {
```

```
            printf(" * ");
```

```
            if (x==2 || x==3 || x==4)
```

```
            {
```

```
                for (i=2; i<=5; i++)
```

```
                {
```

```
                    printf("  ");
```

```
                }
```

```
                printf(" * ");
```

```
                break;
```

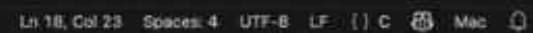
```
            }
```

```
            printf("\n");
```

```
        }
```

```
    return 0;
```

```
}
```



3. Tree angle

```

      *
     * *
    *  *  *
   *   *   *
  *    *    *   *
 *     *     *   *   *

```

```

#include <stdio.h>
int main () {
    int i, k;
    for (i = 1; i <= 5; i++)
    {
        for (k = 1; k <= i; k++)
        {
            printf(" * ");
        }
        printf("\n");
    }
    return 0;
}

```

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Welcome

C: program.c •

C: program.c > main()

```
1  #include <stdio.h>
2  int main()
3  {
4      int k, i;
5      for (i = 1; i <= 5; i++)
6      {
7          for (k = 1; k <= i; k++)
8          {
9              printf("*");
10             }
11             printf("\n");
12         }
13         return 0;
14     }
15
```

PROBLEMS

OUTPUT

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```
macbook@macbooks-MacBook-Pro hello.c % /Users/macbook/hello.c/program
*
**
***
****
*****
macbook@macbooks-MacBook-Pro hello.c %
```

C/C++: ... ✓

/Users/mac...

#### 4. Inverted Triangle

```

* * * *
* * *
* *
*

```

```

#include <stdio.h>
int main()
{
    int i, k;
    for(i=5; i>=1; i--) {
        for(k=i; k>=1; k--)
        {
            printf(" * ");
        }
        printf("\n");
    }
    return 0;
}

```

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Welcome

C\_program.c X

C\_program.c > main()

```
1  #include <stdio.h>
2  int main()
3  {
4      int k, i;
5      for (i = 5; i >= 1; i--)
6      {
7          for (k = i; k >= 1; k--)
8          {
9              printf("%d");
10             }
11             printf("\n");
12         }
13         return 0;
14     }
15
```

PROBLEMS

OUTPUT

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macbook@macbooks-MacBook-Pro hello.c % /Users/macbook/hello.c/program

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\*

macbook@macbooks-MacBook-Pro hello.c %

C/C++: ... ✓

/Users/mac...

5. Pyramid

```

      *
    * * *
  * * * * *
* * * * * * *
* * * * * * * *

```

```

#include <stdio.h>
int main()
{
    int i, k, j, x = 5;
    for (i = 1; i <= 5; i++) {
        for (k = 1; k <= x - i; k++) {
            printf(" ");
        }
        for (j = 1; j <= 2 * i - 1; j++) {
            printf("# ");
        }
        printf("\n");
    }
    return 0;
}

```

EXPLORER

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> TIMELINE

← →

hello.c

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Welcome

C: program.c •

C: program.c > main()

```
1  #include <stdio.h>
2  int main()
3  {
4      int k, j, x = 5, i;
5      for (i = 1; i <= 5; i++)
6      {
7          for (k = 1; k <= x - i; k++)
8          {
9              printf(" ");
10             }
11             |
12             for (j = 1; j <= 2 * i - 1; j++)
13             {
14                 printf("*");
15             }
16             printf("\n");
17         }
18         return 0;
19     }
20
```

PROBLEMS

OUTPUT

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PORTS

macbook@macbooks-MacBook-Pro hello.c % /Users/macbook/hello.c/program

```
*
***
*****
*****
*****
macbook@macbooks-MacBook-Pro hello.c %
```

C/C++: ... ✓

/Users/mac...

## 6. Inverted Pyramid

```

* * * * *
 * * * *
  * * *
   * *
    *
  
```

```

#include <stdio.h>
int main() {
    int i, k, j, x = 1;
    for (i = 5; i >= 1; i--)
    {
        for (k = 1; k <= x; k++)
        {
            printf(" ");
        }
        for (j = 1; j <= 2 * i - 1; j++)
        {
            printf("* ");
        }
        x++;
        printf("\n");
    }
    return 0;
}
  
```

EXPLORER

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Welcome

C: program.c •

C: program.c > main()

```
1  #include <stdio.h>
2  int main()
3  {
4      int k, j, x = 1, i;
5      for (i = 5; i >= 1; i--)
6      {
7          for (k = 1; k <= x; k++)
8          {
9              printf(" ");
10             }
11
12             for (j = 1; j <= 2 * i - 1; j++)
13             {
14                 printf("*");
15             }
16
17             x++;
18             printf("\n");
19         }
20         return 0;
21     }
22
```

PROBLEMS

OUTPUT

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PORTS

```
macbook@macbooks-MacBook-Pro hello.c % ./Users/macbook/hello.c/program
*****
*****
***
*
macbook@macbooks-MacBook-Pro hello.c %
```

C/C++: ... ✓

/Users/mac...

## 7. Half Diamond

```

*
* *
* * *
* * * *
* * * * *
* * * *
* * *
* *
*

```

```

#include <stdio.h>
int main() {
    int i, k, x = 5;
    for (i = 1; i <= 5; i++) {
        for (k = 1; k <= i; k++) {
            printf(" * ");
        }
        printf(" \n");
        x++;
    }
    if (x == 6) {
        for (i = 4; i >= 6; i--) {
            for (k = i; k >= 1; k--) {
                printf(" * ");
            }
            printf(" \n");
        }
    }
    return 0;
}

```



## 8. Diamond



```
#include <stdio.h>
int main()
{
    int k, j, x = 5, i;
    for (i = 1; i <= 5; i++) {
        for (k = 1; k <= x - i; k++) {
            printf(" ");
        }
        for (j = 1; j <= 2 * i - 1; j++) {
            printf("#");
        }
        printf("\n");
        x--;
    }
    if (x == 6) {
        x = 1;
        for (i = 4; i >= 1; i--) {
            for (k = 1; k <= x; k++) {
                printf(" ");
            }

```

```
    }  
    for (j = 1; j <= r * i - 1; j++)  
    {  
        printF(" * ");  
    }  
    printF(" |n");  
    r++;  
    }  
    return 0;  
}
```

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hello.c

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Welcome

C\_program.c X

C\_program.c > main()

```
1  #include <stdio.h>
2  int main()
3  {
4      int k, j, x = 5, i;
5      for (i = 1; i <= 5; i++)
6      {
7          for (k = 1; k <= x - 1; k++)
8          {
9              printf(" ");
10             }
11             for (j = 1; j <= 2 * i - 1; j++)
12             {
13                 printf("x");
14             }
15             printf("\n");
16         }
17         x++;
18         if (x == 6)
19         {
20             x = 1;
21
22             for (i = 4; i >= 1; i--)
23             {
24                 for (k = 1; k <= x; k++)
25                 {
26                     printf(" ");
27                 }
28                 for (j = 1; j <= 2 * i - 1; j++)
29                 {
30                     printf("x");
31                 }
32                 printf("\n");
33                 x++;
34             }
35             return 0;
36         }
37     }
```

PROBLEMS

OUTPUT

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macbook@macbooks-MacBook-Pro: hello.c %

macbook@macbooks-MacBook-Pro: hello.c % /Users/macbook/hello.c/program

x
xxx
xxxxx
xxxxxxx
xxxxxxx
xxxxx
xxx
x

macbook@macbooks-MacBook-Pro: hello.c %

C/C++: ... ✓

/Users/mac...